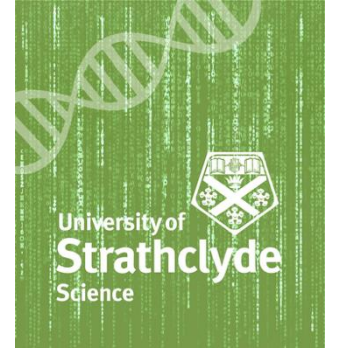


Evolutionary and Structural Analysis of Pathogen Proteins.

Final year UG project 2025-26
2025-11-25 (Week 10)

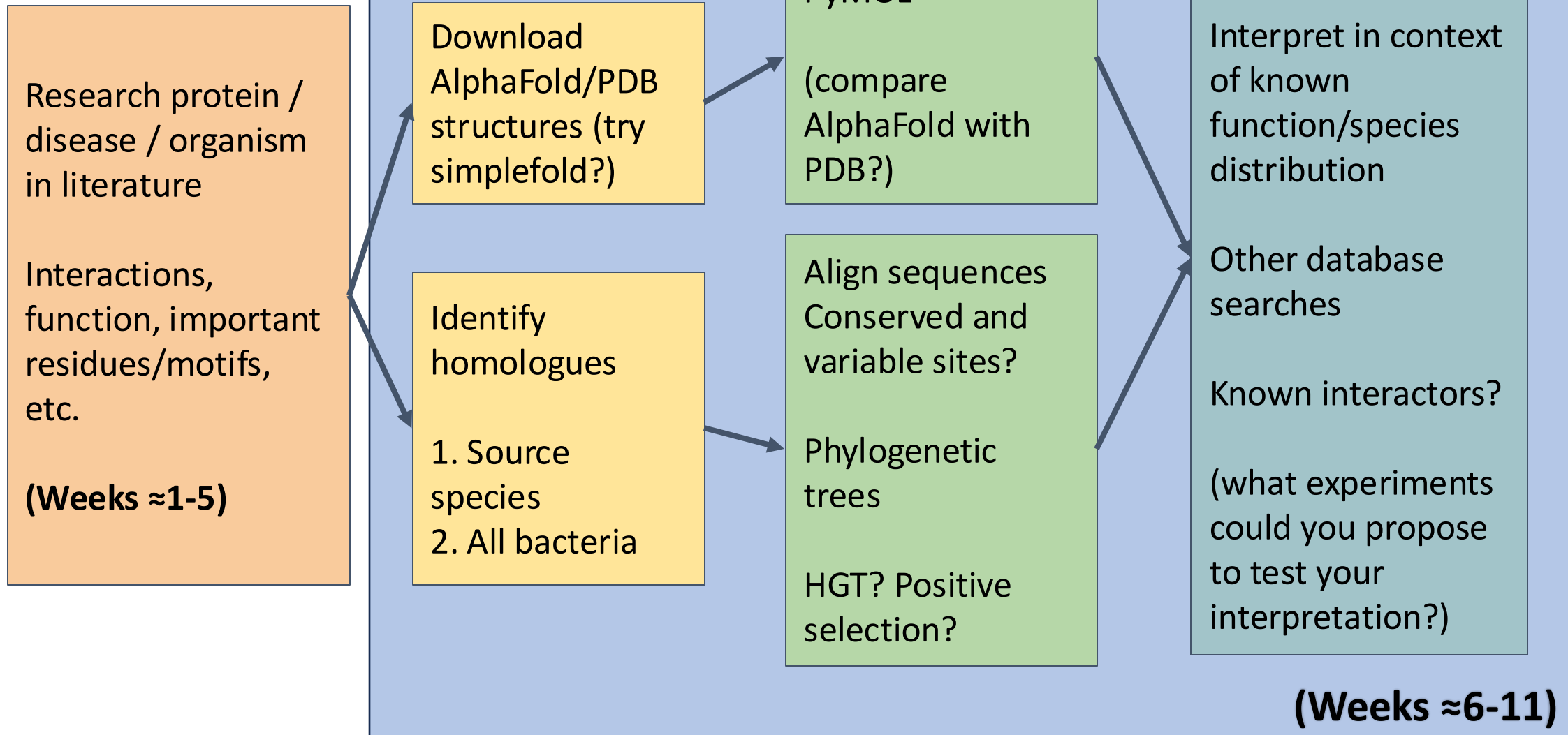
Candidate proteins – start points



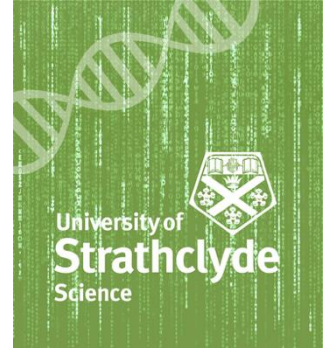
Organism	Host	Gene/Protein	PHI accession	Student
<i>Escherichia coli</i>	<i>Homo sapiens</i>	<i>espY</i>	PHI:8647	LB
<i>Shigella flexneri</i>	<i>Homo sapiens</i>	<i>ipaI</i>	PHI:9253	LT
<i>Candida albicans</i>	<i>Mus musculus</i>	<i>sap6</i>	PHI:10193	IM
<i>Pseudomonas aeruginosa</i>	<i>Homo sapiens</i>	<i>tplE</i>	PHI:6646	AE
<i>Vibrio vulnificus</i>	<i>Mus musculus</i>	<i>vvhA</i>	PHI:6877	JT

<http://www.phi-base.org/>

Workflow



Presentations

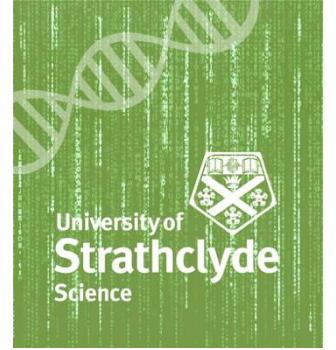


- Presentation dates have been indicated to staff, but not individual times for students
 - Friday 12th December (AM)
 - Monday 15th December (AM/PM)
 - Tuesday 16th December (AM)
- We have been reminded:
 - “Supervisors are not allowed to organise practice sessions for the presentations, but are expected to guide presentation development as part of routine meetings.”
- Old workshop material: <https://sipbs-compbiol.github.io/BM432/>

If I were giving a presentation...

- **Audience:** The markers/assessors
- **Presentation:** $\approx 60\%$ of your mark
 - Cover background information and why the project matters ($\approx 15\%$)
 - Describe aims and hypotheses ($\approx 5\%$)
 - Explain what I did ($\approx 10\%$)
 - Describe my results – 2-3 key items ($\approx 15\%$)
 - Show appropriate data/statistics/figures
 - State how the result relates to the hypothesis/aim in each case
 - Make clear any limitations
 - Describe where the work could go next ($\approx 10\%$)
 - What experiments might be suggested?
 - Style ($\approx 5\%$)

If I were giving a presentation...

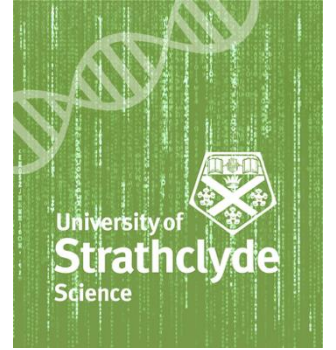


- **Audience:** The markers/assessors
- **On the day:** $\approx 40\%$ of your mark
 - Presentation skills ($\approx 15\%$)
 - Confident, engaging delivery
 - Use slide material well
 - Questions ($\approx 20\%$)
 - Demonstrate understanding of the topic
 - Timing ($\approx 5\%$)
 - Don't run over time
 - Balance time spent on each topic
- **Aim for one slide per minute, approximately**

If I were giving a presentation...

- **Title** (no time)
- **Introduction** (2min): My pathosystem; My protein in my pathosystem
- **Aims** (1min): Use publicly available sequence/structure data to identify/understand variation that corresponds to function/specificity/host range/whatever
- **What I did** (2min): Flowchart of moving from PHI-Base to UniProt/RCSB/AlphaFold etc. through alignment/structure prediction/other to results – with numbers of sequences indicated – **NOT DETAILED METHODS**
- **Results** (4min): Three key outcomes with central figures to talk around, e.g. patterns of sequence conservation/variation and relation to function; potential for multimeric structures/protein-protein interaction; distribution of variants across a phylogeny, etc. – **METHOD DETAIL HERE IF RELEVANT**
- **Future work** (1min): Suggest experiments (wet or dry) that would falsify/test one or more of the described results

If I were giving a presentation...



- **PRACTICE!**
 - In groups if you want/need to
 - Record yourself on your phone
 - Feedback/criticism makes for better talks
- Buy time in questions
 - Repeat the question back to the questioner to let them know you understand it (thinking time!)

Your questions/comments/issues

(What would you like to talk about?)

Next Week's FINAL Group Meeting
Monday 1st December 13:30 HW324

(what doughnuts do you like?)

Useful Links

Useful tools (many others are available)

GalaxyEU: <https://usegalaxy.eu/>

- Sequence alignment (e.g. MAFFT), phylogenetics (e.g. RaxML), positive selection (e.g. codeML)

iTOL: <https://itol.embl.de/>

- Visualisation/annotation of phylogenetic trees

PyMOL: <https://pymol.org/2/> and/or ChimeraX:
<https://www.cgl.ucsf.edu/chimerax/>

- Protein structure visualisation/annotation

Jalview: <http://www.jalview.org/>

- Visualisation of multiple sequence alignments

Useful sites/databases

PHI-base: <http://www.phi-base.org/>

- Proteins involved in host-pathogen interactions, with linked evidence

EMBL AlphaFold: <https://www.alphafold.ebi.ac.uk/>

- AlphaFold predictions for proteins from model organisms

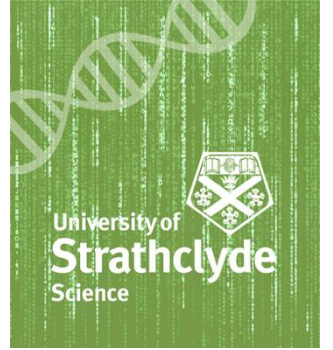
UniProt: <https://www.uniprot.org/>

- Protein sequence (including homologous sequences) and functional information with evidence

RCSB/PDB: <https://www.rcsb.org/>

- Repository of record for protein structures

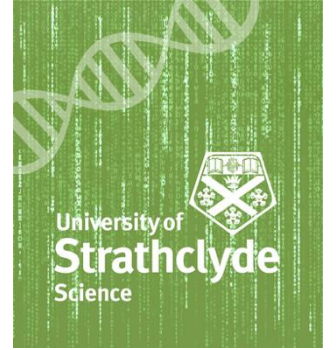
SIPBS CompBiol Sites



- BM432 Project Pages
 - <https://sipbs-compbiol.github.io/bm432-project/>
- An incomplete little book of bioinformatics
 - <https://sipbs-compbiol.github.io/little-bioinformatics-book/>

Project Management Tools

You may want tools to...



- Manage your time
 - E.g. Pomodoro technique (e.g. BeFocused, [Pomofocus](#), [Forest](#))
- Schedule work
 - Reminders (macOS, MS Office)
 - Calendar (macOS, MS Office), with email alerts
 - [Trello](#), [Asana](#), etc.
- Manage your project data and information effectively
 - [How to name files](#)
 - [Project management guidelines](#) (BM432, 2022-23 session; me and Dr Feeney)
 - [How to keep a lab notebook](#)
 - Keeping a computational biology lab notebook:
<https://doi.org/10.1371/journal.pcbi.1004385>
 - [Organising a lab book](#)